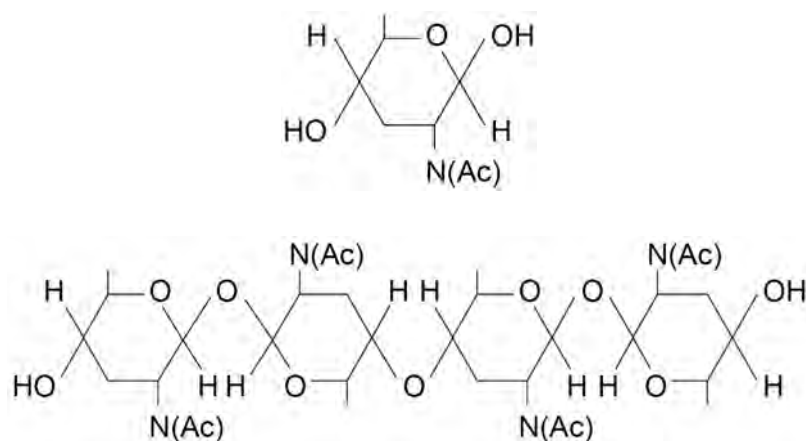


0 2 . 1

Chitin is a polysaccharide. The chitin monomer is a β -glucose molecule with one OH group replaced by an NHCOCH_3 group. NHCOCH_3 can be represented by $\text{N}(\text{Ac})$.

Figure 1 shows the monomer that forms chitin and the chitin polymer.

Figure 1



Chitin has a similar structure to cellulose.

Use **Figure 1** to describe **three** ways the structure of chitin is similar to the structure of cellulose.

[3 marks]

1 _____

2 _____

3 _____

Question 2 continues on the next page

Turn over ►

0 2 . 2

Chitin keeps the tracheae open in the tracheal system of gas exchange in an insect. Gas exchange does **not** occur in the tracheae.

Explain the importance of **one** adaptation of the gas exchange surface in the tracheal system of an insect.

[2 marks]

0 2 . 3

Lignin is a polymer found in the walls of xylem vessels in plants. Lignin keeps the xylem vessel open as a continuous tube.

Explain the importance of the xylem being kept open as a continuous tube.

[3 marks]

Turn over for the next question

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